Xianghang (Scott) Mi

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Research Interests

Cybersecurity, Trustworthy AI & Agent, Security Measurement

Research Overview

The overarching goal of my research is to instill trust into every interaction and communication, both online and offline. To achieve this, my work is organized along two complementary directions: (1) systematically uncovering novel security risks, and (2) designing robust systems and protocols that provide stronger guarantees for security and privacy. In particular, I employ data-driven measurement and ethical attack experiments to identify and characterize security threats, with a focus on emerging cyber-physical infrastructures (e.g., residential proxies, edge CDNs) and new communication paradigms, such as interactions between individuals and AI agents. Building on these insights, I leverage advanced AI techniques and cryptographic primitives to develop and prototype defensive solutions that enhance Internet confidentiality, privacy, and resilience against spam and abuse. Through this dual approach, my research aims to advance the state of the art in trustworthy computing and contribute practical solutions to pressing security challenges.

Education

2015/08 – 2020/06: Indiana University, Bloomington IN
Ph.D. Degree, Department of Computer Science
Advisors: Professor XiaoFeng Wang, and Professor Feng Qian

2009/09 – 2013/06: Beijing Institute of Technology, Beijing, China
B.S. Degree, Department of Software Engineering

Work Experience

12/2025 – Now: Monash University
Assistant Professor (Lecturer)
Department of Software Systems & Cybersecurity

09/2021 – 12/2025: USTC (University of Science and Technology of China)
Pre-Tenure Full Professor
School of Computer Science and Technology

01/2021 – 08/2021: University at Buffalo
Tenure-Track Assistant Professor
Department of Computer Science and Engineering

06/2020 – 05/2021: Parental Leave
Primary caregiver for newborn and family during the COVID-19 pandemic.

06/2019 – 01/2021: Meta, Inc
Research Scientist in Security Infrastructure
Project: Distributed fuzzing platform for automatic vulnerability discovery.
Project: Anti-crawling and bot detection.

05/2014 – 06/2015: Baidu, Inc
Senior Software Engineer in Location-Based Services
Project: Developing the mobile website of Baidu's group buying service.

Honors and Awards

[2025] AsiaCCS 2025 Outstanding PC Member Award

[2022] The Education Grant from the Foundation of Yang YuanQing

[2019] Best Paper Award (3 out of 80) in CSAW Applied Research Competition: This award is given to our paper (SP'19-b) studying the security of virtual personal assistants (i.e., AI agents).

[2019] NDSS Distinguished Paper Award (4 out of 89): This award is given to the paper (NDSS'19) studying the take-down practices of malicious domains.

Datasets and Tools

The following datasets and tools have been released as part of my research projects and are publicly available to the research community.

Datasets

- **RESIP**: Datasets of residential-proxy IPs, comprising millions of distinct IPs observed to serve as residential proxies.
- **ResiFlow**: Dataset of residential-proxy traffic flows, containing over 3TB of traffic data from deployed proxy nodes.
- **SpamHunter**: The largest publicly available dataset of tens of thousands of SMS spam messages.
- **SearchIPT**: Dataset containing over 11 million illicit promotional texts (IPTs) discovered on major search engines.
- **xPIP**: Dataset comprising 12 million distinct posts of illicit promotion (PIPs) published from 580K X accounts.
- **iFTTT**: Comprehensive dataset of services and applets crawled from the iFTTT trigger-action workflow platform.

Tools

- **The RESIP Infiltrator**: Tool for capturing residential-proxy IP addresses from back-connect proxy services.
- **The SpamHunter**: System for continuously discovering SMS spam messages reported by victims on social networks.
- **The IPT Toolchain**: Comprehensive toolchain for capturing and analyzing illicit promotional texts, including hunter, analyzer, and infiltrator modules.
- **The PIP Hunter**: Tool for capturing posts of illicit promotion on social networks, well-tested for the X platform and adaptable to other platforms.

All datasets and tools are available through their respective project websites and GitHub repositories as referenced in my homepage.

Selected Publications

Summary: My research has led to tens of publications in well-recognized computer security venues, e.g., IEEE Security & Privacy, USENIX Security, CCS, NDSS, IMC, CoNEXT, DSN, ACSAC, AsiaCCS, etc. Most publications can be grouped by the following research topics:

- The security risks of residential proxies and reverse proxies: [SP'19-a] [NDSS'21] [CCS'22-a] [EuroSP'25] [arXiv'24-a] .
- The security of AI agents (e.g., voice assistants): [IMC'17] [SP'19-b] [AsiaCCS'24]
- The security of decentralized network infrastructures: [IWQoS'23] [DSN'24] [ACSAC'24]
- Privacy-preserving prevention of harmful content (e.g., spam and malware): [CCS'22-b] [arXiv'24-e] [arXiv'24-f] .
- Detecting and understanding online abuse and cybercrime: [SP'17] [NDSS'18] [NDSS'19] [Security'19] [arXiv'24-c] [WWW'25]

* denotes co-first authors, † stands for corresponding authors.

[1] [WWW'25] Detecting and Understanding the Promotion of Illicit Goods and Services on Twitter
Hongyu Wang, Ying Li*, Ronghong Huang, **Xianghang Mi**†.*
WWW 2025.
Acceptance rate: 20% = 409/2062.

[2] [EuroSP'25] Port Forwarding Services Are Forwarding Security Risks
*Haoyuan Wang, Yue Xue, Xuan Feng, Chao Zhou, **Xianghang Mi**†.*
IEEE European Symposium on Security and Privacy 2025.

[3] [arXiv'24-a] Shining Light into the Tunnel: Understanding and Classifying Network Traffic of Residential Proxies
*Ronghong Huang, Dongfang Zhao, **Xianghang Mi**†, Jingmiao Zhang, XiaoFeng Wang.*
Under submission and in arXiv preprint 2024.

[4] [arXiv'24-b] Reflected Search Poisoning for Illicit Promotion
*Sangyi Wu, Jialong Xue, Shaoxuan Zhou, **Xianghang Mi**†.*
Under submission and in arXiv preprint 2024.

[5] [arXiv'24-c] Fostering Cyber Threat Detection Through Federated Learning
*Yu Bi, Yekai Li, Xuan Feng, **Xianghang Mi**†.*

Under submission and in arXiv preprint 2024.

[6] [arXiv'24-d] SpamDam: Towards Privacy-Preserving and Adversary-Resistant SMS Spam Detection
Yekai Li, Rufan Zhang, Wenxin Rong, ***Xianghang Mi***†.
Under submission and in arXiv preprint 2024.

[7] [ACSAC'24] Dissecting Open Edge Computing Platforms: Ecosystem, Usage, and Security Risks
Yu Bi, Mingshuo Yang*, Yong Fang,* ***Xianghang Mi***†*, Shanqing Guo*†*, Shujun Tang, Haixin Duan.*
ACSAC 2024.
Acceptance rate: 20% = 83/421.

[8] [AsiaCCS'24] Command Hijacking on Voice-Controlled IoT in Amazon Alexa Platform
Wenbo Ding, Song Liao, Long Cheng, ***Xianghang Mi****, Ziming Zhao, Hongxin Hu.*
The 19th ACM Asia Conference on Computer and Communications Security (AsiaCCS 2024).
Acceptance rate: 18% = 418/2,322.

[9] [DSN'24] Stealthy Peers: Understanding Security Risks of WebRTC-based Peer-Assisted Video Streaming
Siyuan Tang, Eihal Alowaisheq, ***Xianghang Mi***†*, Yi Chen, XiaoFeng Wang, Yanzhi Dou.*
IEEE/IFIP DSN 2024.
Acceptance rate: 20% = 42/203.

[10] [IWQoS'23] An Empirical Study of Storj DCS: Ecosystem, Performance, and Security
Hao Li, ***Xianghang Mi***†, Yanzhi Dou, Shanqing Guo†
IEEE/ACM IWQoS 2023.
Acceptance rate: 23.5% = 62/264.

[11] [CCS'22-a] An Extensive Study of Residential Proxies in China
Mingshuo Yang, Yunnan Yu*,* ***Xianghang Mi***†*, Shujun Tang, Shanqing Guo*†*, Yilin Li, Xiaofeng Zheng, Haixin Duan.*
ACM CCS 2022.
Acceptance rate: 22.5% = 218/971.

[12] [CCS'22-b] Clues in Tweets: Twitter-Guided Discovery and Analysis of SMS Spam
Siyuan Tang, ***Xianghang Mi***†*, Ying Li, Xiaofeng Wang, Kai Chen.*
ACM CCS 2022.

Acceptance rate: 22.5% = 218/971.

[13] [NDSS'21] Your Mobile Phone is My Proxy: Understanding and Detecting Mobile Proxy Networks
Xianghang Mi†*, Siyuan Tang, Zhengyi Li, Xiaojing Liao, Feng Qian, Xiaofeng Wang.*
Network and Distributed System Security Symposium 2021.
Acceptance rate: 15.2% = 87/573.

[14] [SP'19-a] Resident Evil: Understanding Residential IP Proxy as a Dark Service
Xianghang Mi*, Xuan Feng, Xiaojing Liao, Baojun Liu, Xiaofeng Wang, Feng Qian, Zhou Li, Sumayah Alrwais, Limin Sun, Ying Liu.*
IEEE Symposium on Security and Privacy 2019.
Acceptance rate: 12% = 84/679.

[15] [SP'19-b] Dangerous Skills: Understanding and Mitigating Security Risks of Voice-Controlled Third-Party Functions on Virtual Personal Assistant Systems
3rd Place (3 out of 80) of CSAW'19 Applied Research Competition
Nan Zhang, ***Xianghang Mi****, Xuan Feng, XiaoFeng Wang, Yuan Tian, Feng Qian.*
IEEE Symposium on Security and Privacy 2019.
Acceptance rate: 12% = 84/679.

[16] [Security'19] Understanding iOS-based Crowdturfing through Hidden UI Analysis
Yeonjoon Lee, Xueqiang Wang*, Kwangwuk Lee, Xiaojing Liao, XiaoFeng Wang, Tongxin Li,* ***Xianghang Mi****.*
USENIX Security Symposium (Security), 2019.
Acceptance rate: 16% = 113/697.

[17] [NDSS'19] Cracking the Wall of Confinement: Understanding and Analyzing Malicious Domain Take-downs
Distinguished Paper Award (4 out of 89)
Eihal Alowaisheq, Peng Wang, Sumayah A Alrwais, Xiaojing Liao, XiaoFeng Wang, Tasneem Alowaisheq, ***Xianghang Mi****, Siyuan Tang, Baojun Liu.*
Network and Distributed System Security Symposium 2019, San Diego, CA.
Acceptance rate: 17% = 89/521.

[18] [NDSS'18] Game of Missuggestions: Semantic Analysis of Search-Autocomplete Manipulations
Peng Wang, ***Xianghang Mi****, Xiaojing Liao, XiaoFeng Wang, Kan Yuan, Feng Qian, and Raheem Beyah.*
Network and Distributed System Security Symposium 2018, San Diego, CA.

Acceptance rate: 21.5% = 71/331.

[19] [Security'17] Picking Up My Tab: Understanding and Mitigating Synchronized Token Lifting and Spending in Mobile Payment
Xiaolong Bai, Zhe Zhou*, XiaoFeng Wang, Zhou Li, **Xianghang Mi**, Nan Zhang, Tongxin Li, S. Hu, Kehuan, Zhang.*
USENIX Security Symposium 2017, VANCOUVER, BC, CANADA.
Acceptance rate: 16.3% = 85/522.

[20] [SP'17] Under the Shadow of Sunshine: Understanding and Detecting BulletProof Hosting on Legitimate Service Provider Networks
*Sumayah Alrwais, Xiaojing Liao, **Xianghang Mi**, Peng Wang, XiaoFeng Wang, Feng Qian, Raheem Beyah, Damon McCoy.*
IEEE Symposium on Security and Privacy 2017, San Jose, CA.
Acceptance rate: 13% = 60/457.

[21] [arXiv'17] Understanding IoT Security Through the Data Crystal Ball: Where We Are Now and Where We Are Going to Be
*Nan Zhang, Soteris Demetriou, **Xianghang Mi**, Wenrui Diao, Kan Yuan, Peiyuan Zong, Feng Qian, XiaoFeng Wang, Kai Chen, Yuan Tian, Carl A Gunter, Kehuan Zhang, Patrick Tague, Yue-Hsun Lin.*
In arXiv preprint 2017.

[22] [IMC'17] An Empirical Characterization of IFTTT: Ecosystem, Usage, and Performance
***Xianghang Mi**, Feng Qian, Ying Zhang, and Xiaofeng Wang.*
ACM Internet Measurement Conference 2017, London, UK.
Acceptance rate: 23.4% = 42/179.

[23] [CoNEXT'16] SMig: Stream Migration Extension For HTTP/2
***Xianghang Mi**, Feng Qian, and XiaoFeng Wang.*
International Conference on emerging Networking EXperiments and Technologies 2016, Irvine, CA.
Acceptance rate: 17.6% = 35/199.

Teaching

2022-2024 Introduction to Cybersecurity, USTC
2022 The Frontier of Cybersecurity Research, Fall 2022, USTC
2021 CSE 489/589 Modern Networking Concepts, Spring 2021,UB
2016-2017 Guest Lectures: CSCI P438 (Computer Networks, IU, Fall 2016), CSCI P538 (Advanced Computer Networks, IU, Fall 2016/Fall 2017).

Professional Service

Conference and Workshop Program Committee: AsiaCCS 2025/2026, WWW 2026, IEEE ICPADS 2023, GenoPri 2020/2021, SKM 2021.

Reviewer for Journals: IEEE Transaction on Mobile Computing 2018, ACM Transactions on Privacy and Security 2021/2022, IEEE Transactions on Dependable and Secure Computing 2021/2022, IEEE Security & Privacy 2021/2022.

Reviewer for Conferences and Workshops: IEEE Security & Privacy 2020, ACM CHI 2020, ACM CCS 2019, IEEE INFOCOM 2019, NDSS 2022/2019/2018, IEEE ICDCS 2017, ACM AsiaCCS 2016, BIGCOM 2016.

Funding Awards

2025 *Blackbox and Zero-Query Evasion Attacks Against LLM-based In-Context Learning*, ¥300K, CCF
2023 *Understanding and Mitigating the Security Risks of Residential Proxies*, ¥300K, NSFC
2022 *Fostering Cyber Threat Detection through Federated Learning*, ¥200K, MSR Asia
2022 *The Innovation Fund for Young Investigators*, ¥90K, USTC

Graduated Students

- Ronghong Huang, MPhil 2025 $\Longrightarrow$ Software Engineer at Tencent
- Yue Xue, MPhil 2025 $\Longrightarrow$ Software Engineer at Agricultural Development Bank of China
- Yong Fang, MPhil 2025 $\Longrightarrow$ Software Engineer at CSNWD Group
- Yu Bi, MPhil 2024 $\Longrightarrow$ Security Engineer at Meituan
- Mingshuo Yang, MPhil 2024 $\Longrightarrow$ Software Engineer at Huawei
- Mingkai Li, BS 2023 $\Longrightarrow$ PhD at Columbia University
- Ying Li, MPhil 2023 $\Longrightarrow$ PhD at UCLA
- Hanliang Jiang, BS 2023 $\Longrightarrow$ Master at UIUC
- Wenxin Rong, BS 2022 $\Longrightarrow$ PhD at University of Delaware